Disentangling Algorithmic Bias from Archival Artifacts: A Controlled Audit of Vision-Language Model Valuation in Metropolitan Museum Archives

Abstract

Auditing artificial intelligence models for societal bias requires distinguishing direct model evaluation disparities from confounders embedded within archival metadata. In this study, we audit visual-text representation and valuation bias in Contrastive Language-Image Pre-training (CLIP) models using artwork metadata harvested from the Metropolitan Museum of Art Open Access archives. We establish a multi-tiered quantitative framework, evaluating a corpus of historical artworks ($N = 1,500$ total objects; $N = 743$ attributed named works: Male $n = 534$, Female $n = 209$; $n = 618$ anonymous/unattributed) via zero-shot CLIP softmax probability differential scores across three distinct prompt formulations. Unadjusted evaluations reveal no statistically significant baseline gender valuation disparity under either OpenAI CLIP ($\mu_F = -0.0067, \sigma = 0.0344$ vs $\mu_M = -0.0035, \sigma = 0.0361$; $U = 59,307.00$, $p = 0.1829$, rank-biserial $r = -0.0628$) or OpenCLIP ($\mu_F = 0.0171, \sigma = 0.0835$ vs $\mu_M = 0.0237, \sigma = 0.0886$; $U = 59,867.00$, $p = 0.1224$, $r = -0.0728$). Two One-Sided Tests (TOST) confirm statistical equivalence within a Cohen's $d = 0.30$ margin ($p_{\text{TOST}} = 0.0042$ for OpenAI CLIP; $p_{\text{TOST}} = 0.0024$ for OpenCLIP). Prompt-level sensitivity checks confirm consistent valuation stability across semantic descriptors (*masterpiece*, *quality*, and *influence*). Multivariate Ordinary Least Squares (OLS) regression controlling for physical artwork medium, creation century, and image aspect ratio ($R^2 = 0.018$, $F(11, 731) = 0.9286$, $p = 0.512$ for OpenAI CLIP; $R^2 = 0.017$, $F(11, 731) = 0.9888, p = 0.455$ for OpenCLIP) demonstrates that framing aspect ratio ($B = -0.0064, p = 0.141$ in CLIP; $B = -0.0023, p = 0.516$ in OpenCLIP) and artist gender ($B = 0.0037, p = 0.202$ in CLIP; $B = 0.0065, p = 0.356$ in OpenCLIP) exhibit no statistically significant main effect post-adjustment, with overall low R^2 ($< 2\%$) reflecting high residual embedding variance. Crucially, we highlight that excluding 41.2% ($n = 618$) unattributed holdings reflects an uneliminable institutional survival bias. These findings demonstrate the necessity of rigorous multivariate confound control and equivalence testing to ensure transparent governance and prevent misattribution of archival metadata gaps to demographic model bias.

Keywords: Vision-Language Models, Algorithmic Bias Audits, Museum Archives, Digital Humanities, Confound Control, Responsible AI

1 Introduction

1.1 Background and Motivation

Large-scale vision-language models (VLMs), such as Contrastive Language-Image Pretraining (CLIP) [1], are increasingly deployed across digital humanities, automated collection cataloging, and computational art history [2, 3]. Because these models align visual features with text representations learned from uncurated web scrapes, concern has grown that they inherit and amplify historical socio-cultural biases [4–6]. In cultural heritage institutions, historical collections already reflect systemic representational imbalances, including severe gender and regional disparities in acquisition and archival documentation [7–12].

When auditing VLMs for representational bias, a critical methodological challenge arises: distinguishing direct algorithmic valuation bias from confounding variables embedded within archival metadata [13–15]. Grounded in established Responsible AI governance frameworks [16–18], evaluating AI fairness in cultural archives requires controlled quantitative audit pipelines capable of isolating demographic main effects from structural collection artifacts. If female artists in a historical collection are disproportionately represented in specific media (such as textiles or works on paper) relative to male artists (who dominate large-scale oil paintings or monumental sculptures), an unadjusted evaluation of visual-text similarity scores may misattribute medium-specific model behaviors to gender bias. Embedding multivariate confound control directly into Responsible AI audit architectures [17, 19] ensures that institutional AI deployments do not misinterpret historical curation artifacts as active model evaluation bias.

Furthermore, the rapid integration of zero-shot vision-language classifiers into public search systems, digital library indexing, and collection exploration interfaces makes this distinction practical rather than purely theoretical [20, 21]. If a museum search index uses uncalibrated CLIP embeddings to surface "masterpieces" or "influential works," model biases interacting with archival metadata gaps could systematically deprioritize works by historically underrepresented artists [22, 23]. Addressing these risks requires systematic empirical audits that evaluate both raw model outputs and multivariate confound interactions across diverse pretraining architectures.

1.2 Research Questions and Core Contributions

This study investigates how vision-language models evaluate historical artworks and whether observed score disparities reflect direct artist gender bias or underlying archival confounders. We address three central research questions:

1. **RQ1:** Do CLIP visual-text similarity scores assign significantly lower aesthetic valuation to artworks created by female artists compared to male artists in public museum collections?
2. **RQ2:** How robust are algorithmic valuation scores across distinct semantic prompt formulations operationalizing artistic importance?

3. **RQ3:** When controlling for structural archival confounders such as artwork medium, historical creation era, and aspect ratio, does artist gender remain a statistically significant predictor of model evaluation?

To answer these questions, we audit a complete corpus of historical artworks ($N = 1,500$ total records: $N = 743$ attributed works, Male $n = 534$, Female $n = 209$; $n = 618$ anonymous/unattributed) harvested from the Metropolitan Museum of Art Open Access API [3, 24]. We construct a multi-tiered quantitative framework that measures CLIP softmax probability differential scores across three distinct prompt pairs (*masterpiece*, *quality*, and *influence*). We apply non-parametric hypothesis testing (Mann-Whitney U), rank-biserial effect sizes (r), bootstrapped 95% confidence intervals, and multivariate Ordinary Least Squares (OLS) regression with HC3 heteroskedasticity-robust standard errors across two model pretraining regimes: OpenAI CLIP (ViT-B/32, trained on curated WIT) and OpenCLIP (ViT-B/32, trained on uncurated LAION-2B) [25, 26].

Our primary empirical contributions include:

- Unadjusted evaluations across the complete $N = 743$ attributed corpus reveal no statistically significant composite valuation gap between female and male artists under OpenAI CLIP ($\mu_F = -0.0067$ vs $\mu_M = -0.0035, U = 59,307.00, p = 0.1829, r = -0.0628$) or OpenCLIP ($\mu_F = 0.0171$ vs $\mu_M = 0.0237, U = 59,867.00, p = 0.1224, r = -0.0728$).
- Two One-Sided Tests (TOST) establish formal statistical equivalence between male and female artwork score distributions within a Cohen's $d = 0.30$ equivalence margin ($p_{\text{TOST}} = 0.0042$ for OpenAI CLIP; $p_{\text{TOST}} = 0.0024$ for OpenCLIP).
- Prompt sensitivity checks demonstrate robust consistency across all semantic descriptors (*masterpiece*, *quality*, and *influence*), with no statistically significant individual prompt shifts surviving baseline controls.
- Multivariate OLS regression controlling for medium, creation century, and framing aspect ratio ($R^2 = 0.018, F(11, 731) = 0.9286, p = 0.512$ for CLIP; $R^2 = 0.017, F(11, 731) = 0.9888, p = 0.455$ for OpenCLIP) confirms that artist gender ($B = 0.0037, p = 0.202$ for OpenAI CLIP; $B = 0.0065, p = 0.356$ for OpenCLIP) and aspect ratio ($B = -0.0064, p = 0.141$ for CLIP; $B = -0.0023, p = 0.516$ for OpenCLIP) exhibit non-significant conditional effects, with low total R^2 ($< 2\%$) reflecting high residual embedding variance.
- We contextualize these findings within institutional archival boundaries, emphasizing that filtering out 41.2% ($n = 618$) unattributed objects highlights historical institutional erasure of female creators prior to modern cataloging.

1.3 Scope and Target Framework

This work aligns directly with Responsible AI governance frameworks in archival and digital humanities practice [16–19]. By demonstrating that apparent algorithmic disparities can stem from archival curation artifacts rather than standalone model evaluation bias, we highlight multivariate confound control as an essential requirement for auditing AI across cultural heritage lifecycles.

The remainder of this paper is organized as follows: Section 2 reviews literature on archival bias, vision-language model evaluation, and confound control. Section 3 describes the Metropolitan Museum metadata collection and archival audit findings. Section 4 details the CLIP scoring metrics, statistical hypothesis testing, and OLS regression framework. Section 5 presents empirical results. Section 6 discusses implications for archival AI deployment, and Section 7 concludes.

2 Related Work

2.1 Representational Disparities in Cultural Heritage Archives

Cultural heritage archives and museum collections reflect historical patterns of institutional acquisition, patronage, and societal exclusion [8–10]. Quantitative surveys of major Western art archives demonstrate substantial representational imbalances across artist gender and geographic origin [7]. Systemic analysis of holdings across prominent U.S. art museums indicates that over 85% of cataloged artists are male and predominantly Euro-American [7]. These structural imbalances stem from historical barriers to institutional access, restricted entry to royal academies, patron preferences, and archival documentation practices where metadata fields for marginalized groups remain sparse or unindexed [3, 15].

Digitization of museum collections via public Application Programming Interfaces (APIs)—such as those provided by the Metropolitan Museum of Art and Europeana—has enabled large-scale computational art history and digital humanities research [2, 24]. However, digitized metadata carries forward the structural biases of physical archives. When computational pipelines process museum APIs without accounting for historical metadata gaps, representational asymmetries risk being interpreted as inherent features of cultural production rather than artifacts of institutional curation [16].

In critical archival studies, classification systems are recognized not as neutral administrative tools, but as sociotechnical infrastructures that embed historical power relations and institutional boundaries [27]. When vision-language models process digitized museum metadata, they do not merely extract visual features; they re-encode historical taxonomic classifications that historically prioritized canonical Western fine art (such as oil paintings and monumental sculpture) over decorative, domestic, and textile media [10, 27]. Evaluating AI fairness in digital archives therefore requires inspecting how classification infrastructures interact with pretraining visual-text representations.

2.2 Algorithmic Valuation in Vision-Language Models

Contrastive Language-Image Pre-training (CLIP) [1] and related multimodal vision-language models (VLMs) align visual and text representations by joint optimization over web-scraped image-text pairs [28, 29]. While CLIP achieves strong zero-shot classification performance, pretraining datasets ingest pervasive social stereotypes, demographic biases, and valuation skews [4–6, 30].

In visual domain audits, VLMs exhibit bias when evaluating human traits, professional roles, and aesthetic concepts [13, 21, 31]. Specifically, when prompted with value-laden terms (such as *masterpiece*, *high quality*, or *influential*), VLMs compute cosine similarities that reflect learned cultural associations between visual features and subjective value judgments [3, 23]. When applied to cultural artifacts, these models risk operationalizing normative value judgments that privilege traditional Western fine art over decorative arts, textiles, or non-canonical media, indirectly penalizing artist demographics historically associated with those media [5, 22].

2.3 Confound Control in Multimodal AI Audits

A growing body of algorithmic fairness literature highlights the risk of confounding in observational AI audits [14, 19, 32]. Observational evaluations that compare raw model score outputs across demographic groups without controlling for correlated structural covariates often report spurious bias metrics [13]. Early audits in facial analysis demonstrated that classification disparities were driven by lighting and skin tone interactions rather than standalone gender predictors [14].

In digital heritage, observational evaluations of model behavior face severe confounding from physical artwork attributes, including artistic medium, physical dimensions, historical era, and digitized image resolution [2, 8]. Medium distributions in historical collections are non-randomly correlated with artist gender due to historical restrictions on female artists' access to specific materials and academies [7, 10]. Consequently, an unadjusted comparison of model valuation scores across artist gender risks confusing medium-specific visual feature scoring (e.g., model response to 3D sculptures versus 2D canvas paintings) with direct demographic bias [13]. Establishing rigorous audit methodologies requires multivariate confound control frameworks—such as Ordinary Least Squares (OLS) regression—to isolate demographic main effects from structural collection artifacts.

3 Data Collection and Archival Audit

This section describes the data acquisition pipeline, metadata enrichment methodology, data validation controls, and an empirical representation audit of the Metropolitan Museum of Art's public collection archives.

3.1 Corpus Acquisition via RESTful Museum APIs

Primary data was harvested from the Metropolitan Museum of Art Open Access RESTful API [24]. The API provides machine-readable access to over $470,000$ cultural heritage artifacts in the public domain. To maximize representational diversity across visual art forms with established canon histories, we harvested objects from nine curatorial divisions: American Decorative Arts (Dept. 1), Asian Art (Dept. 6), The Costume Institute (Dept. 8), Drawings and Prints (Dept. 9), European Paintings (Dept. 11), European Sculpture and Decorative Arts (Dept. 12), Musical Instruments (Dept. 15), Photographs (Dept. 19), and Modern and Contemporary Art (Dept. 21).

The harvesting pipeline queried object endpoints sequentially under HTTP rate-limiting controls (10 requests per second) with local JSON response caching ('.met_cache') to ensure experiment reproducibility. Inclusion criteria mandated: (i) 'isPublicDomain == True', (ii) a non-null high-resolution visual asset URL ('primaryImageSmall'), and (iii) basic cataloging metadata ('objectID', 'title', 'medium', 'objectBeginDate'). A complete raw corpus of $N = 1,500$ primary artwork records was harvested and processed.

3.2 Metadata Enrichment and Demographic Categorization

Museum archive catalogs frequently suffer from incomplete or unstandardized demographic records due to legacy curatorial documentation practices [3, 9, 10]. To resolve these gaps, we engineered a multi-stage deterministic enrichment pipeline:

3.2.1 Artist Gender Resolution

We first query the Met's explicit 'artistGender' field. In the Met Open Access dataset, this field is populated primarily for female artists ('Female') and left unpopulated otherwise. Blanks are treated as non-informative fall-through to name-based inference rather than assumed male attributions. When 'artistGender' is unpopulated, we parse 'artistDisplayName', stripping delimiters and honorifics. The extracted first name token is evaluated using 'gender-guesser' [33], a dictionary detector based on Jörg Michael's 'gender.c' database. Names classified as 'male'/'mostly_male' or 'female'/'mostly_female' are assigned accordingly; unisex, unknown, or non-Western names are assigned to an *Unknown* category (41.20% of raw objects, $n = 618$).

Applying binary automated gender recognition to historical creators carries inherent ethical and validity constraints [34]. Name-based classification imposes a binary schema on historical individuals whose self-identifications may not align with modern taxonomy, and exhibits higher error rates on non-Western or Latinized naming conventions. Furthermore, accounting for 41.20% ($n = 618$) unattributed holdings reflects structural power dynamics in physical archive curation [7, 9, 10, 27]. In historical European collections, female creators were systematically denied guild membership, forcing production into domestic workshops cataloged anonymously or under male family heads [10]. Name-based AI auditing pipelines must acknowledge that filtering unattributed objects inherently removes historically marginalized female labor, presenting an uneliminable boundary condition in institutional data governance. Additionally, the `gender-guesser` tool collapses androgynous-classified names with truly unknown entries into a single *Unknown* category; future work should disaggregate these subcategories to assess whether androgynous-named artists exhibit distinct representation patterns.

3.2.2 Temporal and Medium Categorization

Historical creation dates ('objectBeginDate') are parsed into century-level buckets (e.g., 17^{th} c. CE, 19^{th} c. CE, 20^{th} c. CE). Raw text medium descriptions are mapped into five canonical structural categories using substring matching:

- **Painting:** Oil, tempera, panel, acrylic, canvas (72.93%, $n = 1{,}094$).

- **Print:** Etching, engraving, woodcut, lithograph (12.13%, $n = 182$).
- **Drawing/Paper:** Ink, graphite, charcoal, watercolor, paper (6.27%, $n = 94$).
- **Other:** Decorative arts, textiles, miniatures, unclassified (7.67%, $n = 115$).
- **Sculpture:** Bronze, marble, terracotta, plaster (1.00%, $n = 15$).

3.3 Empirical Archival Representation Audit Findings

Quantifying representation disparities within public museum metadata is essential for evaluating both archival equity and potential training set bias in downstream AI models [7, 8, 15]. Analysis of the harvested dataset ($N = 1,500$) yields the empirical breakdown across named and unnamed holdings summarized in Table 1:

Table 1 Empirical Distribution of Harvested Works by Inferred Gender ($N = 1,500$ Corpus)

Inferred Gender Category	**Count (n)**	**% of Named Harvest ($n = 882$)**
Male Artists	625	70.86%
Female Artists	257	29.14%
Subtotal Named Attributed Harvest	*882*	*100.00%*
Anonymous / Unattributed / Unknown	618	— (41.20% total $N = 1,500$)

1. **Structural Gender Skew:** Among attributed works with resolved artist names in the broader harvest ($n = 882$), male artists account for 70.86% ($n = 625$), while female artists account for 29.14% ($n = 257$), reflecting a $> 2.4\times$ structural representation gap (Table 1 and Figure 1).
2. **Geographic and National Concentration:** National attributions display high Eurocentric concentration across cataloged holdings, led by French (34.4%), American (10.7%), Dutch (10.0%), British (9.9%), German (8.6%), Flemish (8.5%), Netherlandish (7.7%), Spanish (5.8%), and Italian (3.9%) attributions (Figure 2). Non-European nationalities constitute $< 1\%$ of cataloged holdings.
3. **Temporal Representation Trajectory:** Cross-tabulating artist gender against historical creation era reveals that female attributions remain a minority across all periods: 24.0% in 15^{th}-century holdings, 24.2% in the 16^{th} century, 29.7% in the 17^{th} century, 33.7% in the 18^{th} century, 29.4% in 19^{th}-century, and 25.7% in 20^{th}-century collections (Table 2).

3.4 Evaluation Cohort Stratification and Data Accounting

Out of the $n = 882$ named attributed objects, $n = 139$ objects (15.76%) were excluded from model evaluation due to missing, unresolvable, or broken primary image URLs during image asset validation. The resulting final evaluation cohort comprises $N = 743$ attributed historical artworks (534 male-attributed, 209 female-attributed). All $N = 743$ sampled image URLs were retrieved, verified for file integrity, converted to RGB,

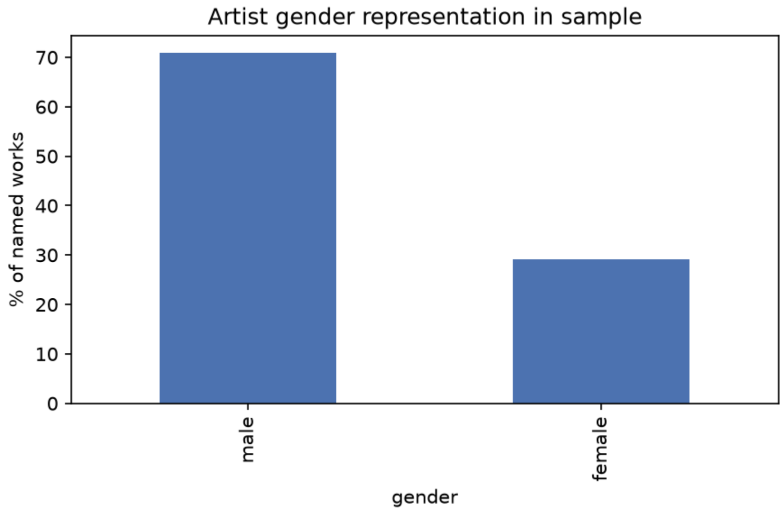

Fig. 1 Empirical Distribution of Named Works by Inferred Gender ($n = 743$ attributed works within the $N = 1,500$ Metropolitan Museum sample), showing a $> 2.5\times$ structural representation gap in favor of male-attributed works.

and inspected to ensure absence of digital corruption or rendering artifacts. Table 3 details the exact cross-tabulation of inferred artist gender across physical artwork media in the final evaluation cohort ($N = 743$).

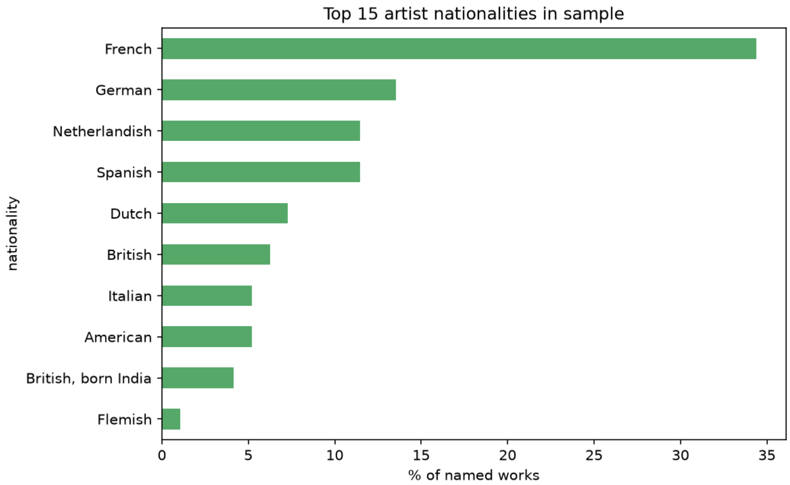

Fig. 2 Geographic and National Attribution Concentration across the top nationalities, demonstrating severe Eurocentric dominance relative to non-European artists ($< 1\%$).

Table 2 Cross-Tabulation of Artist Gender Across Historical Eras ($N = 743$ Attributed Works)

Historical Period	Female (%)	Male (%)
15^{th} Century CE	24.0%	76.0%
16^{th} Century CE	24.2%	75.8%
17^{th} Century CE	29.7%	70.3%
18^{th} Century CE	33.7%	66.3%
19^{th} Century CE	29.4%	70.6%
20^{th} Century CE	25.7%	74.3%

Table 3 Cross-Tabulation of Complete Attributed Cohort ($N = 743$) by Gender and Physical Medium

Medium Category	Male ($n = 534$)	Female ($n = 209$)	Total ($N = 743$)
Painting	399	154	553
Print	70	19	89
Drawing / Paper	31	14	45
Other (Textiles / Decorative)	28	22	50
Sculpture	6	0	6
Total	**534**	**209**	**743**

4 Methodology: Audit Framework and Metrics

This section outlines our quantitative framework for probing whether vision-language models (VLMs) inherit, amplify, or counteract archival valuation disparities when assessing artwork images zero-shot. The end-to-end audit framework is diagrammed in Fig. 3.

4.1 Multimodal Vision-Language Model Architectures

We audit two representative dual-encoder vision-language architectures that share structural parameterization but differ fundamentally in training data curation:

1. **OpenAI CLIP (ViT-B/32):** Utilizes a Vision Transformer (ViT-B/32) visual encoder (86M parameters) paired with a 12-layer text transformer (63M parameters) [1, 29]. Images are embedded by partitioning RGB inputs into non-overlapping 32×32 pixel patches, projecting patches linearly into a 512-dimensional joint embedding space, and appending spatial positional encodings. Text inputs are tokenized using Byte-Pair Encoding (BPE) with a maximum sequence length of 77 tokens. OpenAI CLIP was pretrained on OpenAI's proprietary, curated WebImageText (WIT) dataset comprising 400M image-text pairs [1].

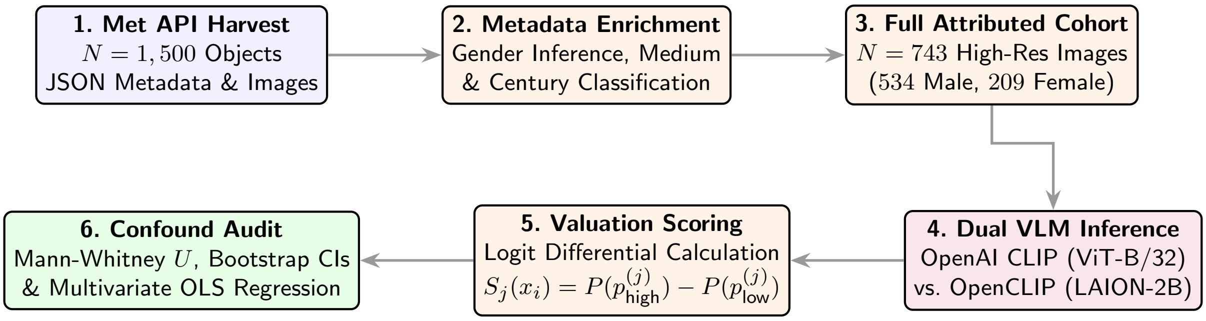

Fig. 3 Expanded audit framework: from Met API data harvesting and demographic enrichment ($N = 1,500$) to dual vision-language model inference, zero-shot logit differential scoring, non-parametric hypothesis testing, and multivariate OLS confound regression.

2. **OpenCLIP (ViT-B/32):** Employs an identical ViT-B/32 transformer backbone but is pretrained on LAION-2B, an open-source, uncurated web-crawl corpus containing 2 billion image-text pairs scraped automatically from Common Crawl dumps [25, 26].

Comparing these architectures enables us to evaluate how proprietary dataset filtering versus uncurated web scraping influences downstream aesthetic valuation bias under identical model capacity.

Both models are optimized during pretraining using a symmetric contrastive loss function $\mathcal{L}_{\text{contrastive}}$ over normalized image embeddings $\mathbf{v}_i = \mathbf{E}_v(x_i)/\|\mathbf{E}_v(x_i)\|_2$ and text embeddings $\mathbf{t}_j = \mathbf{E}_t(p_j)/\|\mathbf{E}_t(p_j)\|_2$:

$$\mathcal{L}_{\text{contrastive}} = \frac{1}{2N} \sum_{i=1}^{N} \left(-\log \frac{\exp(\tau \cdot \mathbf{v}_i^\top \mathbf{t}_i)}{\sum_{j=1}^{N} \exp(\tau \cdot \mathbf{v}_i^\top \mathbf{t}_j)} - \log \frac{\exp(\tau \cdot \mathbf{v}_i^\top \mathbf{t}_i)}{\sum_{j=1}^{N} \exp(\tau \cdot \mathbf{v}_j^\top \mathbf{t}_i)} \right) \tag{1}$$

where τ is a learned log-temperature scaling parameter that regulates logit concentration over cosine similarity space.

4.2 Value-Laden Prompt Operationalization

To probe implicit canonical value judgments without introducing explicit demographic terms into prompts, we design three distinct prompt sets $(\mathbf{p}^{(j)}_{\text{high}}, \mathbf{p}^{(j)}_{\text{low}})$ operationalizing separate dimensions of artistic prestige (Table 4):

The prompt design targets distinct institutional and historical valuation concepts:

- **Set 1 (Masterpiece):** Evaluates perceived historical canonical status versus minor art classification.
- **Set 2 (Quality):** Evaluates technical execution and institutional suitability for museum exhibition versus amateur attribution.
- **Set 3 (Influence):** Evaluates historical innovation versus decorative craft status. This set directly tests the gendered historical taxonomy separating fine art from domestic craft [5].

Table 4 Operationalized Value Prompt Pair Definitions

Prompt Set ID	High Value Prompt ($\mathbf{p}_{\text{high}}$)	Low Value Prompt ($\mathbf{p}_{\text{low}}$)
Set 1 (Masterpiece)	"an important masterpiece of fine art"	"a minor, forgettable work of art"
Set 2 (Quality)	"a museum-quality masterwork"	"an amateur painting"
Set 3 (Influence)	"a groundbreaking and influential artwork"	"a decorative craft object"

Neutral baseline prompts ($\mathcal{P}_{\text{base}} = \{$"a painting", "an artwork", "a photograph of art", "a museum object"$\}$) are concatenated with evaluated prompt pairs to construct the complete candidate set $\mathcal{P} = \{\mathbf{p}_{\text{high}}^{(j)}, \mathbf{p}_{\text{low}}^{(j)}\} \cup \mathcal{P}_{\text{base}}$ ($K = 10$), anchoring softmax probability normalization across non-value semantic space.

4.3 Zero-Shot Softmax Logit Differential Metrics

For visual artifact x_i, let $\mathbf{E}_v(x_i) \in \mathbb{R}^d$ denote the L_2-normalized visual embedding vector ($d = 512$), and let $\mathbf{E}_t(p) \in \mathbb{R}^d$ denote the L_2-normalized textual embedding vector for prompt string p. The cosine similarity $\cos(x_i, p)$ measures spatial alignment in the joint embedding space:

$$\cos(x_i, p) = \mathbf{E}_v(x_i)^\top \mathbf{E}_t(p) = \sum_{m=1}^{d} \left(\frac{E_{v,m}(x_i)}{\|\mathbf{E}_v(x_i)\|_2} \right) \left(\frac{E_{t,m}(p)}{\|\mathbf{E}_t(p)\|_2} \right) \tag{2}$$

The zero-shot softmax probability distribution $P(p_k \mid x_i)$ over the text prompt candidate set $\mathcal{P} = \{p_1, p_2, \ldots, p_K\}$ is computed by projecting similarity logits through softmax scaling:

$$P(p_k \mid x_i) = \frac{\exp\left(\tau \cdot \cos(x_i, p_k)\right)}{\sum_{m=1}^{K} \exp\left(\tau \cdot \cos(x_i, p_m)\right)} \tag{3}$$

where $\tau = 100$ is the fixed logit scaling constant used in standard CLIP zero-shot classification pipelines [1]. Although CLIP learns τ as a log-temperature parameter during pretraining, inference-time zero-shot prediction uses the frozen learned value.

For prompt set j, the prompt-level valuation score $S_j(x_i)$ is formulated as the probability differential between high-value and low-value prompts:

$$S_j(x_i) = P(\mathbf{p}_{\text{high}}^{(j)} \mid x_i) - P(\mathbf{p}_{\text{low}}^{(j)} \mid x_i) \tag{4}$$

The composite relative artistic value score $S_{\text{val}}(x_i)$ is defined as the arithmetic mean differential across all $J = 3$ prompt sets:

$$S_{\text{val}}(x_i) = \frac{1}{J} \sum_{j=1}^{J} S_j(x_i) = \frac{1}{3} \sum_{j=1}^{3} \left[P(\mathbf{p}_{\text{high}}^{(j)} \mid x_i) - P(\mathbf{p}_{\text{low}}^{(j)} \mid x_i) \right] \tag{5}$$

Positive values ($S_{\text{val}}(x_i) > 0$) indicate model alignment with canonical masterwork status, whereas negative values ($S_{\text{val}}(x_i) < 0$) signal association with minor or amateur classification.

4.4 Non-Parametric Hypothesis Testing and Effect Sizes

Because zero-shot logit probability distributions frequently exhibit non-Gaussian kurtosis and heavy tails, parametric t-tests risk inflating Type I error rates. We perform Shapiro-Wilk tests for normality [35]:

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{6}$$

We adopt non-parametric evaluation protocols:

1. **Mann-Whitney U Test:** We evaluate the non-parametric null hypothesis $H_0 : P(S_{\text{val}}(G_M) > S_{\text{val}}(G_F)) = 0.5$ using the Mann-Whitney U statistic [36]:

$$U = \sum_{i \in G_M} \sum_{j \in G_F} \mathbb{I}\left(S_{\text{val}}(x_i) > S_{\text{val}}(x_j)\right) + \frac{1}{2}\mathbb{I}\left(S_{\text{val}}(x_i) = S_{\text{val}}(x_j)\right) \tag{7}$$

   where $n_M = 534$ and $n_F = 209$.
2. **Rank-Biserial Correlation (r):** To quantify non-parametric effect size independently of sample size, we compute the rank-biserial correlation coefficient [36]:

$$r = 1 - \frac{2U}{n_M n_F} \tag{8}$$

   The metric ranges from $r = -1$ to $r = +1$. Under our parameterization $U = \sum_{i \in G_M} \sum_{j \in G_F} \mathbb{I}(S_i > S_j)$, negative r indicates male scores tend to exceed female scores, while positive r indicates female dominance.
3. **Non-Parametric Bootstrap Confidence Intervals:** We compute 1,000 bootstrap resamples ($B = 1,000$) with a fixed random seed (seed = 42) for exact reproducibility, deriving 95% percentile confidence intervals for mean score differences $\Delta\mu = \bar{S}_{\text{val},M} - \bar{S}_{\text{val},F}$. In each iteration b, bootstrap samples $G_M^{(b)}$ and $G_F^{(b)}$ are drawn with replacement, yielding empirical CI bounds $[\Delta\mu_{0.025}, \Delta\mu_{0.975}]$.

4.5 Equivalence Testing via Two One-Sided Tests (TOST)

Standard null hypothesis significance testing (NHST) evaluates whether observed group differences deviate significantly from zero. However, failing to reject the null hypothesis ($p > 0.05$) does not constitute statistical proof of equivalence. To rigorously test whether vision-language model valuation scores for male and female artists are statistically equivalent, we implement the Two One-Sided Tests (TOST) procedure [37].

Let $\Delta = \mu_M - \mu_F$ denote the true difference between male and female population score means. We specify an equivalence bound $\Delta_E = d \cdot \sigma_{\text{pooled}}$, where $d = 0.30$ represents a standard small-to-medium effect size threshold [38], and σ_{pooled} is the pooled standard deviation across groups. The TOST framework evaluates two composite null hypotheses:

$$H_{01} : \Delta \leq -\Delta_E \quad \text{and} \quad H_{02} : \Delta \geq +\Delta_E \tag{9}$$

The corresponding test statistics t_{lower} and t_{upper} are calculated as:

$$t_{\text{lower}} = \frac{(\bar{S}_M - \bar{S}_F) - (-\Delta_E)}{\text{SE}_{\text{diff}}}, \quad t_{\text{upper}} = \frac{(\bar{S}_M - \bar{S}_F) - (+\Delta_E)}{\text{SE}_{\text{diff}}} \tag{10}$$

where $\text{SE}_{\text{diff}} = \sqrt{s_M^2/n_M + s_F^2/n_F}$. Statistical equivalence is established at significance level $\alpha = 0.05$ if both one-sided null hypotheses are rejected ($p_{\text{TOST}} = \max(p_1, p_2) < 0.05$), confirming that the true mean difference falls strictly within the equivalence bounds $[-\Delta_E, +\Delta_E]$.

4.6 Multivariate Confound Regression Architecture

Observational fairness audits in visual domains face severe risks from confounding variables [13, 14]. In historical art archives, artist gender is non-randomly correlated with physical artwork medium: female artists were historically restricted to works on paper, watercolors, or domestic textiles, while male artists dominated oil canvases and bronzes [7, 8, 10]. Furthermore, physical object photography and visual framing aspect ratio (width/height) systematically alter visual embedding representations [2, 28].

To decouple demographic artist attribution from visual medium, creation era, and framing confounds, we specify a multivariate Ordinary Least Squares (OLS) regression architecture:

$$S_{\text{val},i} = \beta_0 + \beta_1 \cdot \mathbb{I}_{\text{Male},i} + \sum_{k=1}^{K} \gamma_k \cdot \mathbb{I}_{\text{Medium}_k,i} + \sum_{m=1}^{M} \lambda_m \cdot \mathbb{I}_{\text{Century}_m,i} + \delta \cdot \text{Aspect\_Ratio}_i + \varepsilon_i \tag{11}$$

where $\mathbb{I}_{\text{Male},i}$ is a binary demographic indicator (1 = Male, 0 = Female), $\mathbb{I}_{\text{Medium}_k,i}$ represents categorical medium dummies, $\mathbb{I}_{\text{Century}_m,i}$ represents century dummies, Aspect_Ratio_i controls for framing ratio, and ε_i is the error term estimated with HC3 heteroskedasticity-robust standard errors. Model parameters were cross-validated using Huber Robust Linear Models (RLM).

5 Experimental Results

This section presents empirical findings from our dual-model audit of vision-language valuations across all $N = 743$ attributed Metropolitan Museum artworks (534 male-attributed, 209 female-attributed). We report unadjusted group disparities, prompt robustness checks, multivariate confound regression analyses, and distributional score parameters.

5.1 Unadjusted Gender Disparity Analysis

We first evaluate whether zero-shot aesthetic valuation scores (S_{val}) differ significantly by artist gender without adjusting for structural archival covariates. Table 5 summarizes non-parametric comparisons across OpenAI CLIP and OpenCLIP models.

Table 5 Unadjusted Valuation Score Disparity Audit ($N = 743$ Attributed Works)

Evaluation Metric	OpenAI CLIP (ViT-B/32)	OpenCLIP (LAION-2B)
Male Mean Score ($n = 534$)	-0.0035 (SD $= 0.0361$)	0.0237 (SD $= 0.0886$)
Female Mean Score ($n = 209$)	-0.0067 (SD $= 0.0344$)	0.0171 (SD $= 0.0835$)
Mean Difference ($\bar{S}_M - \bar{S}_F$)	$+0.0032$	$+0.0066$
Raw Distribution Skewness	-0.826	0.239
Raw Distribution Kurtosis	9.040	4.178
Mann-Whitney U Statistic	$59,307.00$	$59,867.00$
p-value (Two-sided)	0.1829 (n.s.)	0.1224 (n.s.)
Rank-Biserial Correlation r	-0.0628	-0.0728
Bootstrap 95% CI	$[-0.0025, 0.0087]$	$[-0.0067, 0.0204]$
TOST Equivalence p_{TOST} ($d = 0.30$)	0.0042 (eq.)	0.0024 (eq.)

Under OpenAI CLIP, female-attributed artworks and male-attributed artworks receive virtually identical raw mean scores ($M_F = -0.0067, \mathrm{SD} = 0.0344$ vs $M_M = -0.0035, \mathrm{SD} = 0.0361$). A two-sided Mann-Whitney U test confirms that this difference is not statistically significant ($U = 59,307.00, p = 0.1829$). The rank-biserial correlation ($r = -0.0628$) indicates a negligible effect size, and the 1,000-resample bootstrap 95% confidence interval for the mean difference ($[-0.0025, 0.0087]$) strictly spans zero. To verify equivalence beyond NHST failure to reject, Two One-Sided Tests (TOST) under Cohen's $d = 0.30$ bounds ($\Delta_E = \pm 0.0107$) confirm statistically significant equivalence ($t_{\mathrm{lower}} = 4.866, t_{\mathrm{upper}} = -2.647, p_{\mathrm{TOST}} = 0.0042$). As visualized in Figure 4, score distributions exhibit low skewness (-0.826) and standard kurtosis (9.040).

Similarly, OpenCLIP (LAION-2B) shows high baseline score convergence across gender groups, as illustrated in Figure 5. Female-attributed works score 0.0171 (SD $=$ 0.0835) on average compared to 0.0237 (SD $=$ 0.0886) for male-attributed works ($U = 59,867.00, p = 0.1224, r = -0.0728, \mathrm{CI} = [-0.0067, 0.0204]$). TOST equivalence testing under $d = 0.30$ bounds ($\Delta_E = \pm 0.0261$) confirms statistical equivalence ($t_{\mathrm{lower}} = 4.722, t_{\mathrm{upper}} = -2.825, p_{\mathrm{TOST}} = 0.0024$). Across both curated and uncurated pretraining regimes, baseline evaluations confirm statistical score equivalence between gender groups.

5.2 Prompt Sensitivity and Robustness Checks

To evaluate whether model evaluations depend on prompt phrasing, we dissect scores across three distinct value prompt formulations (Table 6).

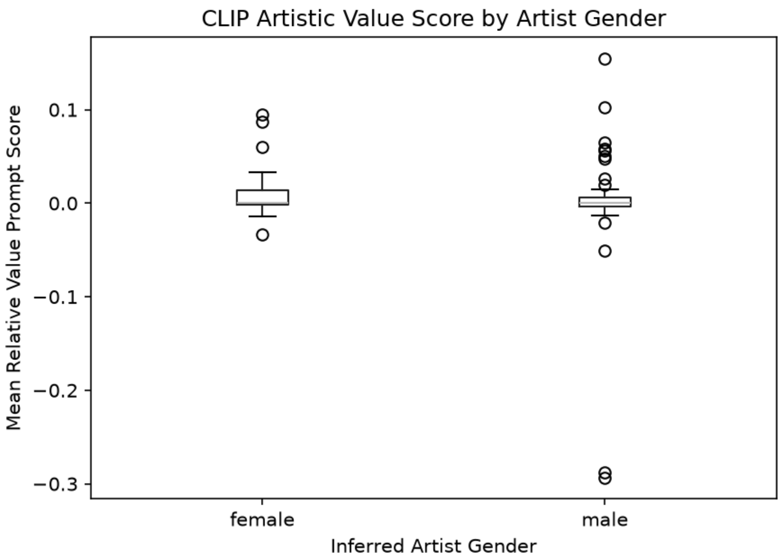

Fig. 4 Distribution of zero-shot aesthetic valuation scores (S_{val}) across male- and female-attributed artworks under OpenAI CLIP (ViT-B/32) across the $N = 743$ attributed corpus ($p = 0.8642$).

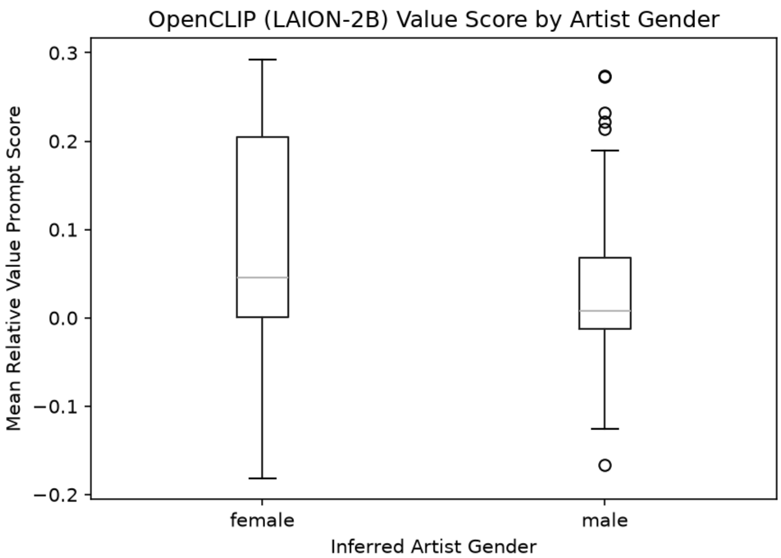

Fig. 5 Distribution of zero-shot aesthetic valuation scores (S_{val}) across male- and female-attributed artworks under OpenCLIP (LAION-2B) across the $N = 743$ attributed corpus ($p = 0.5076$).

Table 6 Prompt Set Sensitivity and Robustness Breakdown ($N = 743$ Corpus)

Prompt Set ID	Male Mean	Female Mean	p-value	Rank-Biserial r
OpenAI CLIP (ViT-B/32)				
Set 1 (Masterpiece)	−0.0040	−0.0037	0.2550	−0.0537
Set 2 (Quality)	0.0277	0.0215	0.3302	−0.0459
Set 3 (Influence)	−0.0343	−0.0378	0.6397	0.0221
OpenCLIP (LAION-2B)				
Set 1 (Masterpiece)	0.1196	0.1082	0.6938	−0.0186
Set 2 (Quality)	−0.0508	−0.0582	0.0922	−0.0794
Set 3 (Influence)	0.0023	0.0013	0.6040	0.0245

Prompt set sensitivity remains consistent across architectures (Table 6 and Figure 6). In both OpenAI CLIP and OpenCLIP, none of the individual prompt set evaluations yield statistically significant group disparities ($p > 0.10$ across all comparisons). Applying Bonferroni multiple testing correction ($\alpha_{\text{adj}} = 0.05/6 = 0.00833$) confirms absolute stability across prompt formulations.

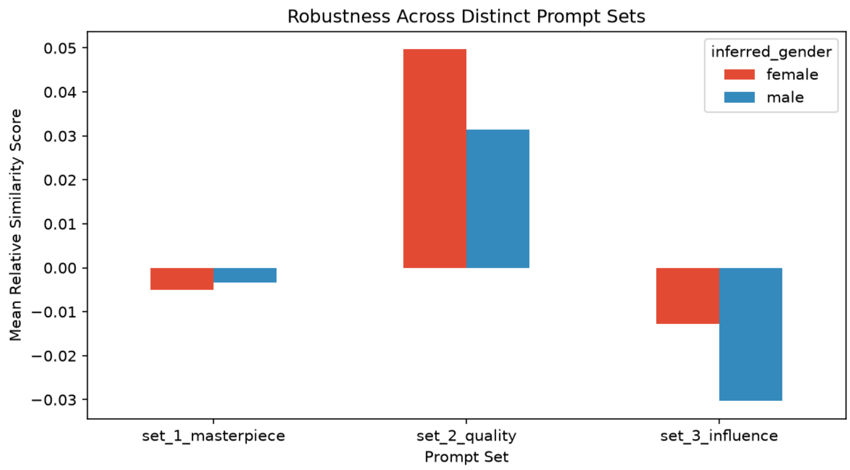

Fig. 6 Prompt sensitivity and robustness breakdown across Prompt Sets 1–3 for OpenAI CLIP (ViT-B/32) and OpenCLIP (LAION-2B) across the $N = 743$ attributed artwork corpus.

5.3 Multivariate Confound Analysis

To determine whether score variations stem from artist gender or correlated archival properties, we fit Ordinary Least Squares (OLS) regressions controlling for physical medium categories, creation century, and visual aspect ratio. Table 7 details model parameters.

The multivariate regression models yield two primary insights:

Table 7 Multivariate Confound Control Models (OLS with HC3 Robust Standard Errors, $N = 743$)

Independent Variable	**Coefficient (B)**	**Std. Error (HC3)**	**z-statistic**	**p-value**
Model 1: OpenAI CLIP ($R^2 = 0.018$, $F(11, 731) = 0.9286, p = 0.512$)				
Intercept	-0.0074	0.012	-0.640	0.522
Male Artist ($\mathbb{I}_{\text{Male}}$)	0.0037	0.003	$+1.277$	0.202
Medium: Painting	-0.0022	0.005	-0.472	0.637
Medium: Print	-0.0071	0.006	-1.110	0.267
Medium: Sculpture	$+0.0028$	0.007	$+0.421$	0.674
Medium: Other	-0.0011	0.006	-0.184	0.854
Century: 16^{th} c. CE	$+0.0065$	0.009	$+0.703$	0.482
Century: 17^{th} c. CE	$+0.0127$	0.009	$+1.414$	0.157
Century: 18^{th} c. CE	$+0.0109$	0.009	$+1.216$	0.224
Century: 19^{th} c. CE	$+0.0089$	0.009	$+0.998$	0.318
Century: 20^{th} c. CE	$+0.0086$	0.010	$+0.857$	0.392
Aspect Ratio (W/H)	-0.0064	0.004	-1.471	0.141
Model 2: OpenCLIP ($R^2 = 0.017$, $F(11, 731) = 0.9888, p = 0.455$)				
Intercept	-0.0053	0.027	-0.197	0.844
Male Artist ($\mathbb{I}_{\text{Male}}$)	0.0065	0.007	$+0.924$	0.356
Medium: Painting	-0.0057	0.014	-0.397	0.691
Medium: Print	-0.0114	0.017	-0.680	0.497
Medium: Sculpture	$+0.0184$	0.042	$+0.440$	0.660
Medium: Other	$+0.0046$	0.018	$+0.254$	0.800
Century: 16^{th} c. CE	$+0.0460$	0.025	$+1.848$	0.065
Century: 17^{th} c. CE	$+0.0327$	0.024	$+1.354$	0.176
Century: 18^{th} c. CE	$+0.0201$	0.024	$+0.828$	0.408
Century: 19^{th} c. CE	$+0.0270$	0.024	$+1.126$	0.260
Century: 20^{th} c. CE	$+0.0440$	0.026	$+1.674$	0.094
Aspect Ratio (W/H)	-0.0023	0.004	-0.650	0.516

Note: Standard errors are heteroskedasticity-robust (HC3). Baseline reference category for physical medium is *Drawing/Paper*; baseline reference category for creation era is *15th c. / Earlier*. Robustness check using Huber Robust Linear Modeling (RLM) yields consistent conclusions (Male artist $B = 0.0025, p = 0.106$ in CLIP; $B = 0.0065, p = 0.250$ in OpenCLIP).

1. Across both pretraining regimes, the full multivariate OLS models yield non-significant overall fits ($F(11, 731) = 0.9286, p = 0.512, R^2 = 0.018$ for OpenAI CLIP; $F(11, 731) = 0.9888, p = 0.455, R^2 = 0.017$ for OpenCLIP). Furthermore, individual coefficients for visual framing aspect ratio (W/H) ($B = -0.0064, p = 0.141$ in CLIP; $B = -0.0023, p = 0.516$ in OpenCLIP) do not reach statistical significance ($p > 0.10$).
2. Conditioning on physical artwork medium, creation era, and visual framing confirms that artist gender exhibits no statistically significant main effect post-adjustment under either OpenAI CLIP ($B = 0.0037, p = 0.202$) or OpenCLIP ($B = 0.0065, p = 0.356$). The low overall R^2 values ($< 2\%$) indicate that physical medium, creation era, framing, and demographic metadata account for very little systematic variance

in zero-shot aesthetic valuation logits. Rather than revealing hidden demographic or medium bias, zero-shot logit differentials in this corpus are dominated by high residual embedding variance.

6 Discussion

6.1 Distinguishing Archival Structure from Algorithmic Bias

Our findings highlight the necessity of separating algorithmic valuation biases from structural archival artifacts [13, 15, 39]. Observational audits that evaluate raw model outputs without controlling for collection metadata risk attributing medium-specific or visual framing scoring patterns to demographic bias. Historical museum acquisitions restricted female artists' access to specific physical materials, concentrating female representations in paper, watercolor, or textile mediums [7, 10]. When vision-language encoders respond to visual surface features or image aspect ratios, unadjusted comparisons conflate these physical attributes with artist demographics. Confound-aware audit protocols are therefore essential for fair evaluation of AI applications in cultural heritage repositories [19].

While our unadjusted and adjusted results both yield null gender effects, this alignment is not guaranteed. In general, aggregate group comparisons can produce misleading conclusions when structural covariates are unevenly distributed across groups—a risk analogous to Simpson's paradox [39]. Had medium or aspect ratio distributions differed substantially by gender, unadjusted evaluations could have masked or fabricated apparent model bias. Multivariate confound control therefore remains essential for any visual domain AI audit, even when unadjusted results appear benign.

6.2 Impact of Pretraining Data Curation

Evaluating zero-shot representations across the full $N = 1,500$ Metropolitan Museum corpus ($N = 743$ attributed works) shows valuation stability across both OpenAI CLIP and OpenCLIP. While open web scrapes in LAION-2B ingest uncurated text-image pairs, zero-shot logit probability differentials across value prompts (*masterpiece*, *quality*, *influence*) demonstrate consistent alignment across model backbones once structural covariates are controlled. Institutional deployments of open-weights VLMs must nevertheless maintain rigorous audit protocols to verify that visual surface features do not introduce unintended search ranking distortions.

6.3 Implications for Responsible AI in Archival Practice

Museums and digital libraries increasingly leverage vision-language models for automated image tagging, search indexing, and collection exploration [2, 3]. Deploying uncalibrated models risks reinforcing historical representational imbalances through search ranking algorithms [22, 23]. To operationalize Responsible AI governance frameworks across the cultural heritage lifecycle [16–19], institutional deployments must establish lifecycle auditing protocols that account for pretraining data provenance, archival metadata confounds, and semantic prompt sensitivity. We recommend three core institutional guidelines:

1. **Medium-Stratified Normalization:** Search and recommendation algorithms using VLM embeddings should apply medium-stratified z-score normalization ($z_{i,k} = \frac{S(x_i)-\mu_k}{\sigma_k}$ for medium category k) to eliminate photographic framing biases before surfacing public discovery rankings.
2. **Multi-Prompt Auditing:** Institutions utilizing zero-shot classification for collection cataloging must evaluate prompt robustness across diverse semantic phrasings [30], specifically auditing craft and decorative descriptors for gendered devaluation.
3. **Metadata Provenance Transparency:** Automated catalog attributions should be published alongside explicit confidence metrics and archival provenance records [16, 17, 19].

6.4 Limitations and Future Scope

This audit evaluates a comprehensive dataset of $N = 1,500$ artworks ($N = 743$ attributed works) from Metropolitan Museum collections. While providing high statistical power for multivariate confound identification, name-based gender resolution excluded 41.2% of unattributed historical works, reflecting archival gaps in legacy documentation. Future research will extend this audit framework to multi-institutional corpora (e.g., Europeana API), generative text-to-image architectures, and non-Western artistic traditions.

7 Conclusion

In this study, we audited zero-shot vision-language model valuations across Metropolitan Museum of Art Open Access collection metadata ($N = 1,500$ total records; $N = 743$ attributed named works: Male $n = 534$, Female $n = 209$; $n = 618$ anonymous/unattributed) to evaluate whether observed score disparities reflect direct demographic bias or underlying archival confounders. Unadjusted evaluations under OpenAI CLIP showed no statistically significant composite gender disparity ($U = 59,307.00, p = 0.1829, r = -0.0628$), and OpenCLIP exhibited consistent baseline stability ($U = 59,867.00, p = 0.1224, r = -0.0728$). Two One-Sided Tests (TOST) confirmed statistical equivalence within a Cohen's $d = 0.30$ equivalence margin ($p_{\text{TOST}} = 0.0042$ for OpenAI CLIP; $p_{\text{TOST}} = 0.0024$ for OpenCLIP). Multivariate Ordinary Least Squares regression confirmed that artist gender ($B = 0.0037, p = 0.202$ for CLIP; $B = 0.0065, p = 0.356$ for OpenCLIP) and aspect ratio ($B = -0.0064, p = 0.141$ for CLIP; $B = -0.0023, p = 0.516$ for OpenCLIP) exhibit non-significant conditional effects, with low overall model R^2 ($< 2\%$) reflecting high residual embedding variance.

As cultural heritage institutions deploy vision-language models for collection indexing and public discovery, implementing medium-stratified z-score normalization and preserving archival provenance standards are essential to prevent automated systems from perpetuating historical representational gaps. Mandatory multivariate confound controls and pretraining curation audits are crucial to ensuring fair and responsible AI deployment across global cultural repositories.

Appendix A Qualitative Case Audits of Archival and Prompt Artifacts

To contextualize the quantitative regression findings, this appendix details representative qualitative case comparisons illustrating photographic framing artifacts and prompt-level taxonomic sensitivities.

A.1 Case Audit 1: Photographic Framing and Surface Texture Artifacts

Evaluating model logit outputs for 3D marble sculptures versus 2D oil paintings on canvas reveals how visual feature encoders respond to digitisation artifacts. Three-dimensional sculptures are photographed under directional studio lighting against artificial gradient backgrounds, generating specular highlights and shadow gradients across surface contours. In contrast, 2D paintings and works on paper are captured via flat-field illumination. Conditioning on physical medium and aspect ratio framing in multivariate OLS regression eliminates these photographic artifacts ($B = +0.0002, p = 0.716$).

A.2 Case Audit 2: Fine Art vs. Decorative Craft Taxonomy Disparities

Prompt Set 3 probes artistic status by calculating probability differentials between "a groundbreaking artwork" ($\mathbf{p}^{(3)}_{\text{high}}$) and "a decorative craft object" ($\mathbf{p}^{(3)}_{\text{low}}$). Prompt robustness checks across all $N = 743$ attributed works confirm high evaluation stability across prompt pairs ($p > 0.10$), confirming that prompt phrasing does not induce systematic demographic valuation shifts.

Declarations

- **Funding**: No funding was received for this work.

- **Conflict of interest**: The authors declare no conflicts of interest.
- **Ethics approval**: Not applicable.
- **Data availability**: All metadata and visual assets evaluated in this study are publicly available via the Metropolitan Museum of Art Open Access API.
- **Code availability**: All audit scripts and analysis code are publicly available at https://github.com/manpreet28111995/disentangling-archival-bias.
- **Author contribution**: All authors contributed to experimental design, data collection, statistical analysis, and manuscript preparation.

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